Penchala Siva Prasanth Kasa

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Available for full-time employment from June 2027 (M.S. completes May 2027).

Summary

Software and ML engineer with 2+ years of production **C/C++** and **Python** experience across backend, ML infrastructure and embedded systems, now an M.S. in CS at UW–Madison. At Qualcomm, used **Python** to benchmark and tune a heterogeneous inference runtime serving **LLM** and vision models. Shipped **ML** scoring at sub-10ms over millions of daily transactions at Visa, and a metering device's software stack at Enphase. Dual degree from IIT Kharagpur; **JEE Mains Paper-II All-India Rank 23**.

Experience

AI Software Intern @ Qualcomm, Inc., *San Diego, USA* May 2026 - Aug. 2026

- Wrote **Python** benchmarking and **model evaluation** harnesses for a heterogeneous **LLM** and vision inference runtime.
- Analyzed cross-device data movement and sync in Python to cut latency; findings drove cost-aware routing of operators.
- Contributed C++ to Qualcomm's ExecuTorch backend, including graph-partitioning logic for operator placement.

Software Engineer @ Enphase Energy, *Bangalore, India | 2 years, full-time* Jul. 2023 - Jul. 2025

- Owned the **software lifecycle** of a new metering device (**C/C++**, Mbed OS/Zephyr), from bootloader to fleet-scale release.
- Cut RAM 98% to 82% and storage by 20% by profiling data paths and redesigning memory layouts and data structures.
- Hardened multithreaded C/C++ services and Protobuf telemetry over CAN/Zigbee; triaged field incidents.
- Scripted build, flash and regression checks in Bash/Python behind CI; drove design docs and code reviews in sprints.

Software Intern @ Visa, Inc., *Bangalore, India* May 2022 - Jul. 2022

- Shipped **ML models** to production: **sub-10ms** scoring over **millions of transactions daily**, with safe fallbacks.
- Automated **model onboarding** with Go/Java microservices over Kafka and Spark, saving 2 weeks per model.
- Centralized scattered YAML model and alert configs into a MySQL schema, improving monitoring and data analysis.

Selected Projects

VecScale - Distributed Vector Search over Embeddings (C++) | *UW-Madison* Jan. 2026 - May 2026

- Built a distributed **vector search** engine: sharded index, parallel top-k workers, MPI, OpenMP and CUDA.
- Scored retrieval with **recall@k** against an exact baseline; scaling and latency runs via CMake and Slurm.

NoFi - Offline Peer-to-Peer Web App | TypeScript, React | MADHACKS, *UW-Madison* Nov. 2025

- Shipped a **TypeScript/React** browser app sending text and voice over audio frequencies (FSK) with no network.
- Wrote the real-time encoder/decoder against the Web Audio API behind a UI state machine that kept the app responsive.

AURA - On-Device Multimodal Assistant | Qualcomm Hackathon, *UW-Madison* Feb. 2026

- Paired vision-language models (CLIP/BLIP) with speech-to-text (Whisper) for real-time on-device scene understanding.
- Served multi-model inference locally on **ONNX Runtime** and the Qualcomm QNN backend with CPU fallback.

Geospatial Real-Time Prediction & Fault-Tolerant Serving | *UW-Madison* Mar. 2024 - Oct. 2024

- Built a real-time serving pipeline (CNN + TabNet) over multi-sensor signals, serving predictions under latency targets.
- Implemented failover so critical paths kept serving through partial outages; live **WebSocket** dashboard and alerting.

Job Board Crawler & Change Detector (Python) | *Personal* 2025 - 2026

- Built a crawler that scans **thousands of postings across 62 job boards** nightly and emails only the new matches.

Technical Skills

- **Languages:** Python, C++, C, Java, Go, SQL, Bash, JavaScript/TypeScript
- **Machine Learning:** PyTorch, TensorFlow, ONNX Runtime, ExecuTorch, scikit-learn, LLM inference, embeddings, recall@k
- **Systems & Backend:** Linux/UNIX, concurrency, memory optimization, REST APIs, microservices, WebSockets, RTOS
- **Cloud & Distributed:** AWS, Azure, Docker, Kafka, Spark, MPI/OpenMP/CUDA, CI/CD
- **Tooling:** Git, CMake/Make, gdb, React, MySQL/SQLite, code review, Agile/Jira

Education

University of Wisconsin, Madison, USA Sep. 2025 - May 2027

Master of Science (M.S.) in Computer Science *School of Computer, Data & Information Sciences*

- **Coursework:** Operating Systems, Machine Learning, HPC, Databases, Computer Vision, AI Agents.
- **Graduate Teaching Assistant** for CS 640: Computer Networks.

Indian Institute of Technology, Kharagpur, India July. 2018 - May 2023

Dual Degree - B.Tech(Hons.) + M.Tech (5-year integrated) *Electronics & Electrical Communication Engineering*

- **Coursework:** ML/AI, Digital Image Processing, Data Structures & Algorithms, Computer Networking.

Achievements

- **Nationally Ranked 23** among 125,000 candidates for IIT JEE Mains Paper-II (2018). (Top 0.018%)
- Nationally Ranked 1318 among 230,000 candidates for IIT JEE Advanced (2018). (Top 0.5%)
- Nationally Ranked 1250 in the Graduate Aptitude Test in Engineering (GATE - 2022). (Top 1%)